AstroLink: Six-Target Reconstruction Report

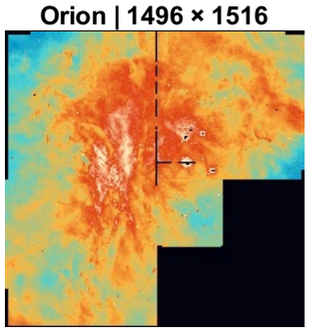
Original and reconstructed science arrays use paired asinh display limits (source p1–p99.7; softening 0.02). Metrics use un-stretched source arrays. Each target is scaled independently; compare each original with its reconstruction, not absolute brightness across targets.

Original (pseudocolour)

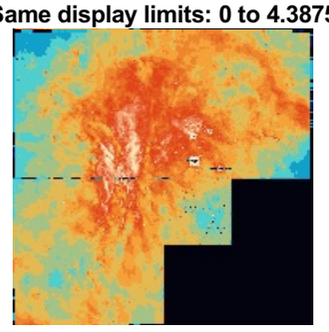
Reconstructed (pseudocolour)

Evaluation metrics

Orion | 1496 × 1516

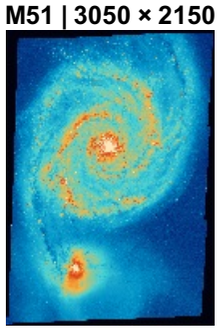


Same display limits: 0 to 4.3875



MSE: 0.0136203
PSNR: 52.675 dB
Global SSIM: 0.996255
Residual SNR: 20.940 dB
BER: N/A (no bitstream/channel)

M51 | 3050 × 2150

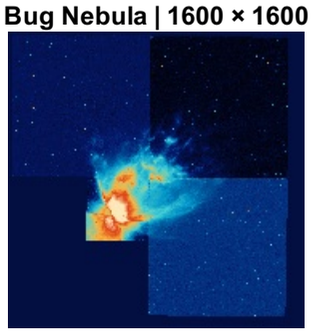


Same display limits: 0 to 0.88398

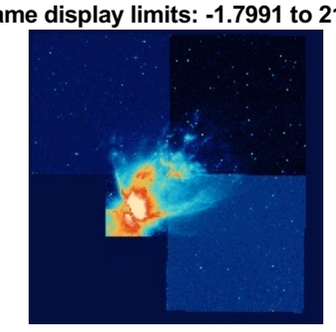


MSE: 0.00577116
PSNR: 63.339 dB
Global SSIM: 0.998769
Residual SNR: 8.173 dB
BER: N/A (no bitstream/channel)

Bug Nebula | 1600 × 1600

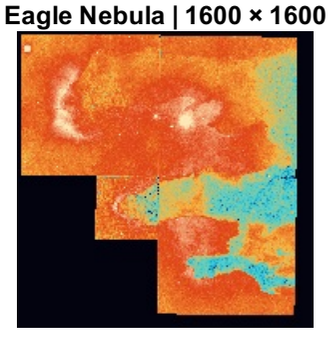


Same display limits: -1.7991 to 215.1

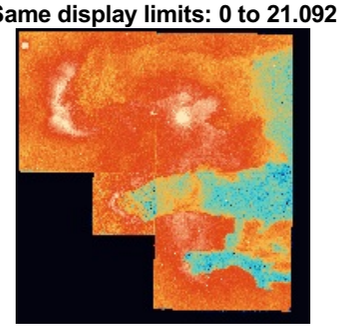


MSE: 0.110549
PSNR: 75.941 dB
Global SSIM: 0.999981
Residual SNR: 39.768 dB
BER: N/A (no bitstream/channel)

Eagle Nebula | 1600 × 1600

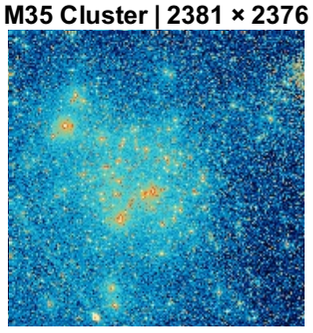


Same display limits: 0 to 21.092

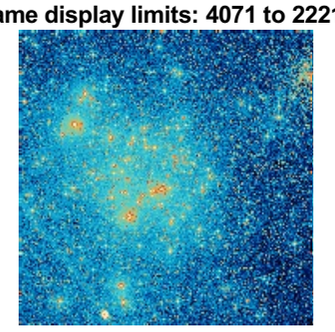


MSE: 0.196382
PSNR: 76.431 dB
Global SSIM: 0.999976
Residual SNR: 31.656 dB
BER: N/A (no bitstream/channel)

M35 Cluster | 2381 × 2376

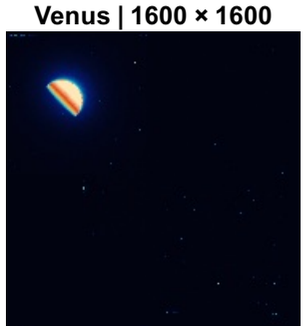


Same display limits: 4071 to 22213

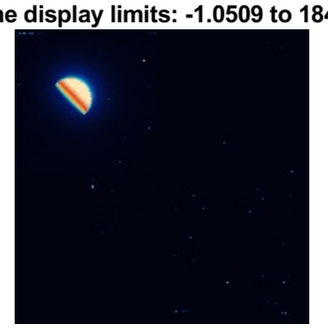


MSE: 0.340003
PSNR: 93.809 dB
Global SSIM: 1.000000
Residual SNR: 79.988 dB
BER: N/A (no bitstream/channel)

Venus | 1600 × 1600



Same display limits: -1.0509 to 1840.5



MSE: 0.120893
PSNR: 92.009 dB
Global SSIM: 0.999999
Residual SNR: 52.733 dB
BER: N/A (no bitstream/channel)